

```

#include <iostream>
#include <cmath>
using namespace std;
double f(double t, double y) {
    return y;
}
double taylorSeriesODE(double t0, double y0, double t, int terms) {
    double sum = y0;
    double t_diff = t - t0;
    for (int n = 1; n <= terms; ++n) {
        double term = pow(t_diff, n) / factorial(n) * f(t0, y0);
        for (int k = 1; k <= n; ++k) {
            term *= f(t0, y0);
        }
        sum += term;
    }
    return sum;
}
unsigned long long factorial(int n) {
    unsigned long long result = 1;
    for (int i = 1; i <= n; ++i) {
        result *= i;
    }
    return result;
}
int main() {
    double t0 = 0;
    double y0 = 1;
    double t;
    int terms;
    cout << "Enter the time (t) at which to estimate y: ";
    cin >> t;
    cout << "Enter the number of terms in the Taylor series: ";
    cin >> terms;
    double result = taylorSeriesODE(t0, y0, t, terms);
    cout << "The approximate solution at t = " << t << " is: " << result << endl;
    return 0;
}

```